

Ainesh Sootha

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Experience

Amazon Web Services - Bedrock AI

Apr.2024 – Present

Software Engineer (ML)

Seattle, WA

Bedrock Generative AI

- Engineered performance optimizations for production-scale open source LLMs (Llama, Deepseek, Mistral) on AWS Bedrock, translating novel research into scalable solutions through collaboration with research scientists.
- Built high-performance distributed services using Python, Rust, Java, and Kubernetes.
- Experimented with a wide range of novel techniques including distributed caching, distributed inference, and speculative decoding to achieve measurable performance improvements. This involved a lot of rapid prototyping and extensive benchmarks (and automation thereof).
- Worked with and optimized multiple inference frameworks (vLLM, sgLang, tensorRT-LLM, NVIDIA Dynamo) while developing custom solutions for enterprise-scale deployments
- Designed and implemented a scalable distributed service for LLM hosting and lifecycle management, enabling seamless import and deployment of custom fine-tuned models into AWS Bedrock infrastructure.
- Partnered directly with a diverse set of external customers ranging from teams of ML engineers to specialized research groups, delivering hyper-optimized model deployments (on AWS Bedrock Custom Model Import) tailored to specific use cases and performance requirements.

Amazon

Jul.2023 – Apr.2024

Software Engineer

Seattle, WA

Just Walk Out Technology – Auth Orchestration

- Worked on large scale, low-latency software for orchestration of customer entry authentication and authorization. (Java, Kotlin)
- Dealt with complex Payments, Fraud & Abuse systems while building a high-throughput, Tier-1 service.

Amazon

May.2022 – Aug.2022

Software Engineer Intern

Seattle, WA

Just Walk Out Technology – Customer Identification & Tracking

- Built a trajectory visualization web application that accurately mapped customer journeys across stores. This helped reduce store deployment time and cost.
- Developed a KPI viewing application that pulled data from different sources and helped Science and Research teams in JWO visualize model performance.

State Street Corp

Jun.2021 – Aug.2021

Software Engineer Intern

New York, NY

- Implemented a custom PDF generation service for **BestX** TCA dynamic web reports using chromium and Node.

Projects and other work (github.com/aineshsootha)

Teaching Assistant - Software Engineering tools (Python)

Realistic Ocean Simulation - GPU-parallelized generation of realistic Ocean Waves (Tessendorf) (C++, OpenGL)

Purdue: Data Mine – Viasat (Java, Kafka), **EPICS** – Habitat for Humanity

Education & Skills

Purdue University

2019 – 2023

BS, **Computer Engineering** + Minor in Philosophy

West Lafayette, IN

- **Relevant Courses:** Data Structures, Computer Graphics, OS, OOP in C++, Compilers, Networks, Linear Algebra